

AIR POLLUTION AND CPCB STANDARDS

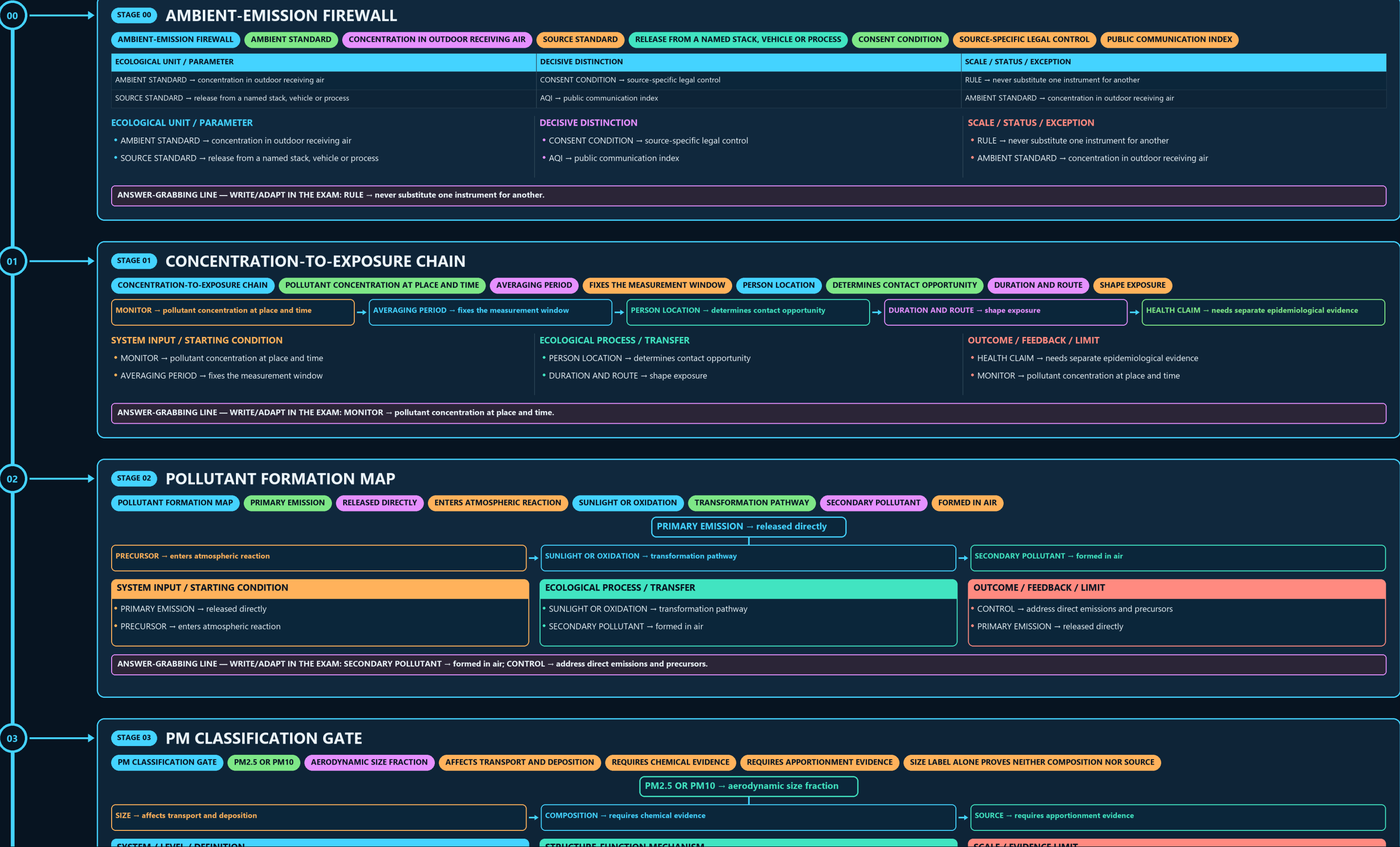
AMBIENT-EMISSION FIREWALL → CONCENTRATION-TO-EXPOSURE CHAIN → POLLUTANT FORMATION MAP → PM CLASSIFICATION GATE → SMOG AND ACID-DEPOSITION MATRIX → NAAQS DATA GATE

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • active generation g2 • explicit approval of this exact generation is required • all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles



PRECURSOR → enters atmospheric reaction

SECONDARY POLLUTANT → formed in air

PRIMARY POLLUTANT → released directly

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SECONDARY POLLUTANT → formed in air; CONTROL → address direct emissions and precursors.

03

STAGE 03

PM CLASSIFICATION GATE

PM CLASSIFICATION GATE

PM2.5 OR PM10

AERODYNAMIC SIZE FRACTION

AFFECTS TRANSPORT AND DEPOSITION

REQUIRES CHEMICAL EVIDENCE

REQUIRES APPORTIONMENT EVIDENCE

SIZE LABEL ALONE PROVES NEITHER COMPOSITION NOR SOURCE

PM2.5 OR PM10 → aerodynamic size fraction

SIZE → affects transport and deposition

COMPOSITION → requires chemical evidence

SOURCE → requires apportionment evidence

SYSTEM / LEVEL / DEFINITION

PM2.5 OR PM10 → aerodynamic size fraction

SIZE → affects transport and deposition

STRUCTURE-FUNCTION MECHANISM

COMPOSITION → requires chemical evidence

SOURCE → requires apportionment evidence

SCALE / EVIDENCE LIMIT

TRAP → size label alone proves neither composition nor source

PM2.5 OR PM10 → aerodynamic size fraction

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TRAP → size label alone proves neither composition nor source.

04

STAGE 04

SMOG AND ACID-DEPOSITION MATRIX

SMOG AND ACID-DEPOSITION MATRIX

PRECURSOR EMISSIONS PLUS SUNLIGHT-DRIVEN CHEMISTRY

SURFACE OZONE

SECONDARY OXIDANT

ACID DEPOSITION

SULPHUR AND NITROGEN PRECURSOR PATHWAY

WET OR DRY

DEPOSITION ROUTES

| ECOLOGICAL UNIT / PARAMETER                               | DECISIVE DISTINCTION                                     | SCALE / STATUS / EXCEPTION                                |
|-----------------------------------------------------------|----------------------------------------------------------|-----------------------------------------------------------|
| SMOG → precursor emissions plus sunlight-driven chemistry | ACID DEPOSITION → sulphur and nitrogen precursor pathway | RULE → precursor is not the final atmospheric product     |
| SURFACE OZONE → secondary oxidant                         | WET OR DRY → deposition routes                           | SMOG → precursor emissions plus sunlight-driven chemistry |

ECOLOGICAL UNIT / PARAMETER

SMOG → precursor emissions plus sunlight-driven chemistry

SURFACE OZONE → secondary oxidant

DECISIVE DISTINCTION

ACID DEPOSITION → sulphur and nitrogen precursor pathway

WET OR DRY → deposition routes

SCALE / STATUS / EXCEPTION

RULE → precursor is not the final atmospheric product

SMOG → precursor emissions plus sunlight-driven chemistry

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → precursor is not the final atmospheric product.

05

STAGE 05

NAAQS DATA GATE

NAAQS DATA GATE

IDENTIFY EXACT SUBSTANCE OR FRACTION

PRESERVE THE NOTIFIED UNIT

AVERAGING PERIOD

ANNUAL, DAILY OR SHORTER ONLY IF SOURCED

AREA AND COMPLIANCE RULE

READ NOTIFICATION

NO TABLE

POLLUTANT → identify exact substance or fraction

UNIT → preserve the notified unit

AVERAGING PERIOD → annual, daily or shorter only if sourced

AREA AND COMPLIANCE RULE → read notification

SYSTEM / LEVEL / DEFINITION

POLLUTANT → identify exact substance or fraction

UNIT → preserve the notified unit

STRUCTURE-FUNCTION MECHANISM

AVERAGING PERIOD → annual, daily or shorter only if sourced

AREA AND COMPLIANCE RULE → read notification

SCALE / EVIDENCE LIMIT

NO TABLE → no invented limit

POLLUTANT → identify exact substance or fraction

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: AVERAGING PERIOD → annual, daily or shorter only if sourced.

06

STAGE 06

AQI STATUTORY BOUNDARY

AQI STATUTORY BOUNDARY

MONITORED POLLUTANT CONCENTRATIONS

PUBLIC-FACING CATEGORY

NOTIFIED AMBIENT STANDARD

SOURCE LIMIT

SEPARATE EMISSION CONTROL

AQI CATEGORY IS NOT EITHER STANDARD TABLE

| ECOLOGICAL UNIT / PARAMETER                | DECISIVE DISTINCTION                     | SCALE / STATUS / EXCEPTION                          |
|--------------------------------------------|------------------------------------------|-----------------------------------------------------|
| INPUT → monitored pollutant concentrations | NAAQS → notified ambient standard        | VERDICT → AQI category is not either standard table |
| OUTPUT → public-facing category            | SOURCE LIMIT → separate emission control | INPUT → monitored pollutant concentrations          |

ECOLOGICAL UNIT / PARAMETER

INPUT → monitored pollutant concentrations

DECISIVE DISTINCTION

NAAQS → notified ambient standard

SCALE / STATUS / EXCEPTION

VERDICT → AQI category is not either standard table

AIR POLLUTION AND CPCB STANDARDS • TILE 2/4 • same master rows 2533-5767 of 10834 • continues below

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: AVERAGING PERIOD → annual, daily or shorter only if sourced.

06

STAGE 06 AQI STATUTORY BOUNDARY

AQI STATUTORY BOUNDARY

MONITORED POLLUTANT CONCENTRATIONS

PUBLIC-FACING CATEGORY

NOTIFIED AMBIENT STANDARD

SOURCE LIMIT

SEPARATE EMISSION CONTROL

AQI CATEGORY IS NOT EITHER STANDARD TABLE

ECOLOGICAL UNIT / PARAMETER

INPUT → monitored pollutant concentrations

OUTPUT → public-facing category

ECOLOGICAL UNIT / PARAMETER

• INPUT → monitored pollutant concentrations

• OUTPUT → public-facing category

DECISIVE DISTINCTION

NAAQS → notified ambient standard

SOURCE LIMIT → separate emission control

DECISIVE DISTINCTION

• NAAQS → notified ambient standard

• SOURCE LIMIT → separate emission control

SCALE / STATUS / EXCEPTION

VERDICT → AQI category is not either standard table

INPUT → monitored pollutant concentrations

SCALE / STATUS / EXCEPTION

• VERDICT → AQI category is not either standard table

• INPUT → monitored pollutant concentrations

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: VERDICT → AQI category is not either standard table.

07

STAGE 07 INSTITUTION AND CLEARANCE LADDER

INSTITUTION AND CLEARANCE LADDER

NATIONAL COORDINATION AND FRAMEWORK

SPCB OR PCC

CONSENT, MONITORING AND ENFORCEMENT

CTE OR CTO

SOURCE-OPERATION PERMISSION LAYER

ENVIRONMENTAL CLEARANCE

PRIOR PROJECT-APPRAISAL LAYER

CPCB → national coordination and framework

SPCB OR PCC → consent, monitoring and enforcement

CTE OR CTO → source-operation permission layer

ENVIRONMENTAL CLEARANCE → prior project-appraisal layer

SYSTEM / LEVEL / DEFINITION

• CPCB → national coordination and framework

• SPCB OR PCC → consent, monitoring and enforcement

STRUCTURE-FUNCTION MECHANISM

• CTE OR CTO → source-operation permission layer

• ENVIRONMENTAL CLEARANCE → prior project-appraisal layer

SCALE / EVIDENCE LIMIT

• RULE → approvals remain legally distinct

• CPCB → national coordination and framework

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SPCB OR PCC → consent, monitoring and enforcement.

08

STAGE 08 MONITORING EVIDENCE LADDER

MONITORING EVIDENCE LADDER

ONE LOCATION

SPATIAL COVERAGE

DATA COMPLETENESS

USABLE TIME RECORD

DISPERSION AND ACCUMULATION CONTEXT

SCALE ONLY AS FAR AS EVIDENCE SUPPORTS

STATION → one location

NETWORK → spatial coverage

DATA COMPLETENESS → usable time record

METEOROLOGY → dispersion and accumulation context

SYSTEM / LEVEL / DEFINITION

• STATION → one location

• NETWORK → spatial coverage

STRUCTURE-FUNCTION MECHANISM

• DATA COMPLETENESS → usable time record

• METEOROLOGY → dispersion and accumulation context

SCALE / EVIDENCE LIMIT

• INFERENCE → scale only as far as evidence supports

• STATION → one location

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: INFERENCE → scale only as far as evidence supports.

09

STAGE 09 SOURCE-APPORTIONMENT CUBE

SOURCE-APPORTIONMENT CUBE

SOURCE MIX CHANGES BY LOCATION

ACTIVITY AND METEOROLOGY CHANGE

PM, OZONE AND GASES HAVE DIFFERENT PATHWAYS

MODEL AND SAMPLING ASSUMPTIONS MATTER

NO UNIVERSAL SHARE

DATE AND SCOPE EVERY ESTIMATE

ECOLOGICAL UNIT / PARAMETER

CITY → source mix changes by location

SEASON → activity and meteorology change

DECISIVE DISTINCTION

POLLUTANT → PM, ozone and gases have different pathways

METHOD → model and sampling assumptions matter

SCALE / STATUS / EXCEPTION

NO UNIVERSAL SHARE → date and scope every estimate

CITY → source mix changes by location

ECOLOGICAL UNIT / PARAMETER

• CITY → source mix changes by location

• SEASON → activity and meteorology change

DECISIVE DISTINCTION

• POLLUTANT → PM, ozone and gases have different pathways

• METHOD → model and sampling assumptions matter

SCALE / STATUS / EXCEPTION

• NO UNIVERSAL SHARE → date and scope every estimate

• CITY → source mix changes by location

09

STAGE 09 SOURCE-APPORTIONMENT CUBE

SOURCE-APPORTIONMENT CUBE SOURCE MIX CHANGES BY LOCATION ACTIVITY AND METEOROLOGY CHANGE PM, OZONE AND GASES HAVE DIFFERENT PATHWAYS MODEL AND SAMPLING ASSUMPTIONS MATTER NO UNIVERSAL SHARE DATE AND SCOPE EVERY ESTIMATE

| ECOLOGICAL UNIT / PARAMETER                                                                                                                                        | DECISIVE DISTINCTION                                                                                                                                                                | SCALE / STATUS / EXCEPTION                                                                                                                                                  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CITY → source mix changes by location                                                                                                                              | POLLUTANT → PM, ozone and gases have different pathways                                                                                                                             | NO UNIVERSAL SHARE → date and scope every estimate                                                                                                                          |
| SEASON → activity and meteorology change                                                                                                                           | METHOD → model and sampling assumptions matter                                                                                                                                      | CITY → source mix changes by location                                                                                                                                       |
| ECOLOGICAL UNIT / PARAMETER <ul style="list-style-type: none"><li>CITY → source mix changes by location</li><li>SEASON → activity and meteorology change</li></ul> | DECISIVE DISTINCTION <ul style="list-style-type: none"><li>POLLUTANT → PM, ozone and gases have different pathways</li><li>METHOD → model and sampling assumptions matter</li></ul> | SCALE / STATUS / EXCEPTION <ul style="list-style-type: none"><li>NO UNIVERSAL SHARE → date and scope every estimate</li><li>CITY → source mix changes by location</li></ul> |
| ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: NO UNIVERSAL SHARE → date and scope every estimate.                                                                |                                                                                                                                                                                     |                                                                                                                                                                             |

10

STAGE 10 PROGRAMME-RESPONSE ARCHITECTURE

PROGRAMME-RESPONSE ARCHITECTURE LONG-RUN STRUCTURAL PLANNING STANDARDS-BASED CLASSIFICATION OVER A PERIOD EPISODIC GRADED RESPONSE STATUTORY DELHI-NCR REGIONAL COORDINATION NO SUBSTITUTION EACH SERVES A DIFFERENT FUNCTION

| ECOLOGICAL UNIT / PARAMETER                                                                                                                                                           | DECISIVE DISTINCTION                                                                                                                                          | SCALE / STATUS / EXCEPTION                                                                                                                                                |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NCAP → long-run structural planning                                                                                                                                                   | GRAP → episodic graded response                                                                                                                               | NO SUBSTITUTION → each serves a different function                                                                                                                        |
| NON-ATTAINMENT → standards-based classification over a period                                                                                                                         | CAQM → statutory Delhi-NCR regional coordination                                                                                                              | NCAP → long-run structural planning                                                                                                                                       |
| ECOLOGICAL UNIT / PARAMETER <ul style="list-style-type: none"><li>NCAP → long-run structural planning</li><li>NON-ATTAINMENT → standards-based classification over a period</li></ul> | DECISIVE DISTINCTION <ul style="list-style-type: none"><li>GRAP → episodic graded response</li><li>CAQM → statutory Delhi-NCR regional coordination</li></ul> | SCALE / STATUS / EXCEPTION <ul style="list-style-type: none"><li>NO SUBSTITUTION → each serves a different function</li><li>NCAP → long-run structural planning</li></ul> |
| ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: NO SUBSTITUTION → each serves a different function.                                                                                   |                                                                                                                                                               |                                                                                                                                                                           |

11

STAGE 11 EVIDENCE AND ANSWER SPINE

EVIDENCE AND ANSWER SPINE AMBIENT, EMISSION, CONCENTRATION AND EXPOSURE SOURCE, PRECURSOR, ATMOSPHERIC FORMATION AND RECEPTOR NAAQS, AQI, CONSENT, NCAP, GRAP AND CAQM STATION, AVERAGING PERIOD AND APPORTIONMENT BOUNDARY TARGET, ATTAINMENT, CURRENT DATA AND PYQ KEY STAY DISTINCT

DEFINE → ambient, emission, concentration and exposure

TRACE → source, precursor, atmospheric formation and receptor

GOVERN → NAAQS, AQI, consent, NCAP, GRAP and CAQM

MEASURE → station, averaging period and apportionment boundary

AUDIT → target, attainment, current data and PYQ key stay distinct

SYSTEM / LEVEL / DEFINITION

- DEFINE → ambient, emission, concentration and exposure
- TRACE → source, precursor, atmospheric formation and receptor

STRUCTURE-FUNCTION MECHANISM

- GOVERN → NAAQS, AQI, consent, NCAP, GRAP and CAQM
- MEASURE → station, averaging period and apportionment boundary

SCALE / EVIDENCE LIMIT

- AUDIT → target, attainment, current data and PYQ key stay distinct
- DEFINE → ambient, emission, concentration and exposure

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: AUDIT → target, attainment, current data and PYQ key stay distinct.

E

EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT EFFECTIVE AIR-POLLUTION POLICY DEPENDS ON ACCURATE SOURCE APPORTIONMENT (SCIENTIFICALLY) COMMON ANALYTICAL ERROR TO AVOID TREATING CROP-RESIDUE BURNING AS THE DOMINANT NCAP SET CITY-SPECIFIC PARTICULATE-MATTER (PM2.5/PM10) REDUCTION TARGETS FOR IDENTIFIED IMPLEMENTATION CRITIQUE PROGRESS TRACKING UNDER NCAP HAS FACED CHALLENGES BALANCED FRAMING

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OPTIONAL ECOLOGICAL PROCESS DEPTH <ul style="list-style-type: none"><li>CAUTION: Effective air-pollution policy depends on accurate source apportionment (scientifically)</li><li>CAUTION: Common analytical error to avoid: treating crop-residue burning as the dominant,</li><li>NCAP set city-specific particulate-matter (PM2.5/PM10) reduction targets for identified</li><li>CAUTION: Implementation critique: progress tracking under NCAP has faced challenges</li></ul> | SCALE, PARAMETER OR STATUS LIMIT <ul style="list-style-type: none"><li>CAUTION: Balanced framing: NCAP represents a scientifically sound target-setting framework;</li><li>CAUTION: Real-time AQI readings depend on monitoring-station density and placement; sparse</li><li>CAUTION: NAAQS permissible limits are themselves periodically reviewed against evolving health</li><li>CAUTION: Attributing health outcomes (mortality, morbidity) precisely to a specific pollution</li></ul> | COMPARATIVE RESTORATION NUANCE <ul style="list-style-type: none"><li>The Commission for Air Quality Management in NCR and Adjoining Areas Act, 2021</li><li>GRAP is a reactive, severity-triggered emergency-response mechanism; NCAP is a</li><li>Source apportionment studies show the relative contribution of crop-residue burning to</li><li>CAQM was established partly because prior coordination on Delhi-NCR air pollution had</li></ul> |
| OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                   |